NOVA IMS
Information Management School

MARCH, 2025

WHO-LIFE

A GLOBAL INITIATIVE TO UNDERSTAND LIFESTYLES WORLDWIDE

HANDOUT
MACHINE LEARNING 2024/2025



01 I. INTRODUCTION

In an era of rapid globalization, urbanization, and shifting societal values, understanding how people structure their daily lives has become more critical than ever. Recognizing the importance of these insights, the World Health Organization (WHO) has launched WHO-LIFE, a global research initiative aimed at analyzing and categorizing **lifestyle patterns** across different countries and regions.

From the wellness-driven routines of Scandinavian countries to the fast-paced urban lifestyles of major metropolitan hubs, people's choices are influenced by health, economics, culture, and social structures. But what are the dominant lifestyle patterns worldwide? And how do factors like income, work-life balance and financial planning shape how people live?

II. HANDOUT GOALS

The primary components and goals of the project are:

1. **Preprocessing and Exploratory Data Analysis (EDA):** Conduct a through data exploration to understand the dataset's main characteristics and prepare the data for effective modeling.
2. **Additional Insights:** True innovation often comes from exploring beyond predefined methods. Think critically and creatively by investigating aspects of the dataset that could lead to **meaningful discoveries with real impact.** You are given the flexibility to choose your own approach, as long as the analysis provides **valuable insights into global lifestyles**. This could include:
    a. *Comparing lifestyle trends across the world*—Are there lifestyles that are predominant in certain regions/continents? What trends can you find?
    b. *Investigating citizen behaviors*—Do eco-conscious individuals exhibit different spending or wellness habits? Is there a link between financial stability and a fitness-conscious lifestyle?
    c. *New data* — Integrate your data with data from other sources to generate new insights.
3. **Descriptive Modeling:** In this phase, you will **apply unsupervised learning to uncover natural lifestyle segments** within the population. Using clustering techniques, the objective is to identify and characterize distinct groups based on behaviors and characteristics, effectively defining different lifestyle choices. Your focus is on segmenting the population, analyzing the key features that define each group, and understanding what differentiates them. **For this Handout, you are expected to present an initial approach to the segmentation**.

02 III. DATASET

You have access to one dataset:

- *world_citizens.csv*: Containing the data you will use for the clustering phase. This corresponds to your **training set dataset**, where you will find detailed information about each citizen but **does not contain the target feature (this column will only be made available after the Handout delivery).**

Complete information about the data is available in the Project File.

IV. DELIVERABLES

Upon the project's deadline, you will be required to submit:

- A *Jupyter* **notebook (or zip file) containing all the code** used throughout the project **and any extra data you may have used**. Make sure to include detailed markdown cells and comments within the notebook to guide readers through the code. Clearly explain the purpose of each code segment, the insights gained, and the decisions made.
    - Name your file in the format ***ML_GroupXX_Handout,*** where ***GroupXX*** should be your group number.

1. Deliveries after the deadline will be penalized at 1 point per day.
2. Deliveries made before the deadline will receive a bonus of 0.1 points per day of delivery in advance (up to a maximum of 0.5 point).

Deadline: April 24th at 18h00

03 VI. EVALUATION

Your work will be evaluated according to the following criteria:

CRITERIA	PERCENTAGE (%)	MAXIMUM GRADE (OUT OF 20)
Notebook Quality	5	1
Exploratory Data Analysis (EDA)	25	5
Preprocessing	30	6
Additional Insight	25	5
Descriptive Modeling	15	3

Your grade will reflect our assessment of the quality of your work in terms of quality of writing, clarity, conciseness, correctness and efficiency. Please find below more details about what is taken for each topic:

- **Notebook Quality:** Assess notebook clarity and readability, including whether the goals are clearly stated, the code contains comments, the techniques are appropriate, the insights and conclusions are highlighted, and the implications for further steps.
- **Exploratory Data Analysis (EDA):** Describe the data and extract meaningful insights that may be helpful in addressing the problem in hand.
- **Preprocessing:** This stage includes all the steps from raw data into data ready for modeling: data cleaning, transformation, and reduction. It also entails context-related transformations of the input features, creating new features, and accompanying explanations. Here, we evaluated the quality of your implementations, its justifications, and insights extracted at this stage.
- **Additional Insight**: Describe your strategy for the additional insight's objective. This section is separated into different components:
  - Formulation and adequacy
  - Difficulty
  - Correctness/efficiency of implementation
  - Explanation and discussion of results
- **Descriptive Modeling:** Describe your **initial strategy** for the clustering objective. This section is separated into different components:
  - Modelling approach – implementation and reasoning behind any clustering model used. You should explain the feature selection for the models.
  - Description of resulting clusters- Each cluster should be statistically and visually explored and described, emphasizing the characteristics that differentiate them.